Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/12/2005
PI Name:	Pearlman, Howard G. Ph.D.		
Project Title:	The Cool Flames Space-Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Howard Pearlman was formerly in the database as a PI at Drexel University. This is the same Howard Pearlman. He is now at GRC working with John McQuillen in PS.		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	09/01/2003	**End Date:**	09/18/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1006		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	This grant considers high Lewis number premixed gas flames and cool flames at reduced gravity. The Cool Flames Experiment successfully passed the science concept review in 8/00 and preparation is in progress for RDR. Description: This is a two part project. The first part aims to better understand the effect of Lewis number on the diffusive-thermal instability in premixed gas flames. In essence, the structure and stability of premixed flames is determined by the chemistry coupled with transport of heat and mass (chemical species). When the ratio of the rate of heat diffusion to mass diffusion is greater than one (i.e., a high Lewis number), the flame becomes unstable and develops spatio-temporal patterns that appear in the form of target and spiral waves. These flame instabilities were predicted in the late 1970's, but have only recently been observed in the lab, albeit highly distorted due to complications associated with buoyant convection. At microgravity, experiments will eliminate buoyant complexities and isolate the roles of heat and mass diffusion on the flame structure and stability, the system will be reduced to a tractable experiment and it will be modelled. This effort is a ground-based study. The second part of the project aims to study cool flames, which are very weak flames that occur in almost all fuel-air mixtures when they are heated. For over thirty years, cool flames have been oberseved in closed reactors, yet complications associated with natural convection have prevented researchers from modelling these systems and quantitatively comparing experimental data. At microgravity, however, a unique environment exists where complications associated with natural convection can be mitigated and the structure, stability and evolution of cool flames can be determined. This effort encompasses two distinct investigations with complimentary but different objectives. The first effort is directed at developing a greater understanding of the high-Lewis number diffusive-thermal instability. This is a ground-based effort. The role of Lewis number, gas mixtures, heat loss and the role of additives/inhibitors are studied in both normal and reduced gravity. The system in which this instability will be investigated is a large diameter tube filled with a mixture of known gases that are ignited at one end. The large diameter tube is important because the instability manifests itself as a long wavelength wave on the surface of an otherwise freely-propagating flame as the flame traverses the tube. The buoyant rise of hot products due to gravity scales as the square root of g (earth’s gravity) multiplied by the tube diameter. To eliminate the complexities associated with the buoyant transport processes, reduced gravity testing must be conducted to obtain an unobscured view of the competing diffusive processes of mass and heat. The second effort is aimed at understanding low-temperature oxidation reactions and cool flames. This is a flight-definition program. It will focus on closed, static reaction vessels into which a mixture of known, premixed gases is introduced. The chamber and thus the gases are heated until a reaction, often a precursor to ignition, is initiated. In these systems, buoyancy alters the hydrodynamic, thermal and concentration fields. In turn, this alters the flame shape and reaction rate. Since the time required for reaction initiation can be on the order of minutes, hours or as long as days, these experiments must be performed in long duration reduced gravity facilities. In addition, all existing cool flame and thermal ignition models (e.g., Semenov, Frank-Kamenentskii, Gray-Yang) neglect natural convection and assume conduction heat transfer. While this is not a valid assumption for most unstirred, normal gravity cool flames and autoignitions, it is exclusively valid in unstirred reactors at microgravity.The current objective for this portion of the project is to provide technical direction to the engineering teams designing and fabricating the breadboards of critical systems prior to the Requirement Definition Review. Instabilities similar to the diffusive-thermal instability to be studied have been observed in combustion synthesis of materials, as well as a myriad of other chemical and biological systems, including electrical propagation of signals in and around heart tissue (note, breakup of spatially organized waves is believed to be responsible for heart problems, most notably sudden cardiac death). In material synthesis, burning of a solid is involved, which mandates that the combustion can only be understood on the surface of the material. Since the current investigation involves gaseous components, transport throughout the system can be observed and parameters more easily varied. The knowledge thus gained, which is not available in any other configuration, should be directly applicable to combustion synthesis of materials as well as a providing a fundamental understanding of the couplings between reaction and diffusion, common to all combustion systems. The cool flame investigation is aimed at understanding processes that occur leading to spontaneous ignition. As such, the investigation will lead to an increased understanding of the physics behind knock in engines, which fuel manufacturers avoid by using additives, together with safety related concerns in spacecraft. Such an understanding is lacking in current combustion theory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Refined chemical kinetic models can be baselined using the data obtained from this experimental program. These refined mdoels can then be used to more accurately predict the combustion behavior of hydrocarbon fuels at low temperatures. It is relevant to autoignition in newly developed Homogeneous Charge Compression Ignition engines and also useful to mitigate potential fire hazards in space-based platforms.
	NUMERICAL PROGRESS Numerical analyses of an extended Gray-Yang reactive-diffusive-convective model have been performed to clarify the role of buoyant convection on the temperature, species and velocity profiles associated with cool flames in closed reactors. New results have been obtained at Earth, Lunar, Martian and Zero-gravity. Qualitatively, the behavior of the reaction is similar to that calculated without inclusion of a branching step, e.g., the Sal'nikov model. These results will be published in a peer-reviewed combustion journal.2. Numerical analyses of an extended Gray-Yang reactive-diffusive-convective model have been performed to clarify the role of buoyant convection on the temperature, species and velocity profiles associated with cool flames in closed reactors. New results have been obtained at Earth, Lunar, Martian and zero-gravity. Qualitatively, the behavior of the reaction is similar to that calculated without inclusion of a branching step, e.g., the Sal'nikov model. These results are being prepared for publication in a peer-reviewed combustion journal (e.g., Combustion and Flame). A skeletal chemical kinetic mechanism has also been developed by Wang and Mou to predict cool flame behavior in a CSTR (continuously stirred tank reactor) has been modified to include diffusive transport, a requirement for all unstirred, static reduced gravity studies. The model has been numerically coded and preliminary results have been obtained. For the first time, time-periodic, damped temperature oscillations are predicted at reduced-gravity. Experimentally, such temporal oscillations have been not (as yet) been observed at reduced-gravity for unity Lewis number within the available 20s on the KC135 aircraft, yet have been observed at high Lewis number (Le>>1) in butane-oxygen mixtures that have significant helium dilution. The oscillatory nature of cool flames at reduced-gravity is intricately linked with the conductive heat versus diffusive species transport. Additional experiments and numerical analysis is planned to further clarify the thermokinetic and diffusive couplings and explore the way these couplings lead to oscillations, sustained or

Task Progress:	damped. EXPERIMENTAL PROGRESS Low pressure experiments using equimolar propane: oxygen and propane:oxygen:nitrogen premixtures were conducted during the August, 2004 KC-135 aircraft flight week at NASA GRC. Approximately 40 tests were conducted at four different temperatures ranging from 320 to 350C using a static, unstirred, quartz reactor. Radially-propagating, diffusive-controlled single cool flames, multi- and single-stage ignitions were observed following an induction period that is dependant on the mixture composition, temperature, pressure and gravity level. Using the CIR LLL-UV camera, spectral imaging of the light emission, owed to excited formaldehyde, was performed. Never before seen images of propane:oxygen cool flames at microgravity were obtained. Post-analysis of the spectral data will clarify the spatio-temporal formaldehyde concentration, which will be used to validate recently developed detailed low-temperature kinetic models of cool flames (Fairlie, Griffiths, Hughes, Pearlman, 2004). With refined kinetic models, an improved predictive capability will be acquired for possible use in a variety of industrial processes including auto-ignition in diesel engines and currently developed low-emission HCCI (homogeneous charge compression ignition) engines such as those under development at Sandia National Labs and by automotive manufacturers. Additional KC-135 aircraft testing is required to obtain data over a broader range of temperatures, pressures, and mixture compositions to improve our understanding of auto-ignition, help mitigate fire safety hazards in spacecraft, learn new ways to use cool flames to refine petroleum fuels and generate hydrogen gas for fuel cells (a new effort is being explored independently by our group at Drexel, DOE, and NASA GRC). Representative cool flame experiments have also been conducted at Martian and Lunar gravity are shown below to demonstrate the importance of natural convection on the flame structure, temperature and species profiles. An understanding of these effects is important to mitigate fire hazards and optimize the combustion process of hydrocarbon fuels at different g-levels, including those associated with Mars and the Moon, in support of the President's new exploration vision. Note also that the existing cool flame apparatus can also be used to support the President's Exploration vision to study partial oxidation of hydrocarbons as well as carbon monoxide oxidation (the primary Martian-derived fuel) at partial-gravity (Lunar and Matian) conditions as demonstrated in KC135 tests. Knowing the manner in which convection and diffusion affect the heat release and loss rates in these systems, thus temperature of the reacting gas, the amount of heat, power and electricity (e.g., thermoelectrically-derived) that can be derived from reacting CO on Mars and/or the Moon can be determined. CONFERENCE PAPER A technical paper entitled "Cool flames in space" was accepted and presented at 30th International Combustion Symposium held at UIC in July, 2004. OUTREACH An overview of NASA's Martian rover flights has been introduced in the Compressible Gas Dynamics course (MEM 320) taught by Professor Pearlman at Drexel University. NASA-relevant examples such as temperature, pressure and density changes on leading edges of supersonic aircraft are used to demonstrate the principles of compressible gases. Cool flame research conducted aboard NASA's KC135 aircraft and reduced-gravity facilities has also been introduced.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Papers from Meeting Proceedings	Fairlie, R., Griffiths, J., Hughes, K., Pearlman, H. "Cool flames in space: experimental and numerical studies of propane combustion" 30th Symposium (International) on Combustion, Chicago, IL (July, 2004), Gas-phase Combustion Jul-2004
Presentation	Pearlman, H., Foster, M. "The Cool Flames Experiment: Recent Results at Reduced and Partial Gravity" 42nd Aerospace Sciences Meeting Jan-2004